

R Training Guide: Using ARD with the {cards} Package

Shu Guo

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1. Introduction to ARD (Analysis Results Dataset)

The **Analysis Results Dataset (ARD)** is part of the CDISC **Analysis Results Standard (ARS)** framework, which aims to standardize how statistical analysis results are structured, stored, and reused.

Traditionally, clinical trial results (tables, figures, and listings) are generated as static outputs (e.g., PDFs or RTFs), which are difficult to reuse, trace, or automate. The ARS framework addresses these limitations by introducing a machine-readable, structured representation of analysis results and their metadata.

Within this framework: - **ARD** represents the *analysis results data itself* (e.g., summary statistics, p-values, counts) - **ARM (Analysis Results Metadata)** describes how those results were produced (methods, populations, variables)

An ARD serves as a “single source of truth” for analysis outputs, enabling:

- **Reproducibility:** Re-run calculations seamlessly across different execution environments.
- **Traceability:** Trace results directly back to source data (e.g., ADaM datasets like ADSL or ADAE).
- **Reusability:** Feed the same result dataset into multiple outputs (tables, figures, reports, interactive apps).
- **Automation:** Streamline reporting workflows by decoupling statistical computation from display formatting.

In practice, an ARD is typically a **tidy dataset** where each row represents a result (e.g., statistic for a group), along with sufficient context (variables, groupings, methods) to reproduce and interpret that result.

The {cards} R package provides a convenient way to **generate ARD-like structured outputs**, making it easier to integrate ARD principles into reproducible R workflows.

1.1 ARD Data Structure and ARM (Analysis Results Metadata)

ARD Data Structure

An **Analysis Results Dataset (ARD)** is a structured, machine-readable dataset where each row represents a **single analysis result** along with the necessary context to interpret it.

The ARD is typically organized in a tidy format, meaning:

- **Each row:** Represents one result (e.g., a mean, count, or p-value).
- **Each column:** Defines a specific attribute describing that result (e.g., variable, group level, statistic name).

Key Components of an ARD

An ARD should generally include the following types of information:

1. Analysis Context

- **Analysis variable** (e.g., AVAL, PARAM)
- **Population** (e.g., safety, ITT)
- **Analysis set flags** (e.g., SAFFL, ITTFL)
- **Visit / time point** (e.g., AVISIT, AVISITN)

2. Grouping Variables

- Treatment groups (e.g., TRT01A)
- Subgroups (e.g., sex, age group, parameter, analysis visits)

These define *how results are stratified*.

3. Result Type and Statistic

- Statistic name (e.g., MEAN, N, SD, P_VALUE)
- Unrounded Numeric result (e.g., AVAL, CHG, or similar)

4. Additional Metadata

- Units (e.g., mg/dL)
- Denominator (for percentages)
- Method (e.g., t-test, ANOVA)

Example ARD Structure

group1	group1_level	group2	group2_level	stat_name	stat_level	variable	stat
TRT01A	Placebo	PARAM	Weight	MEAN	Mean	AVAL	72.3
TRT01A	Placebo	PARAM	Weight	SD	Std Dev	AVAL	10.2
TRT01A	Placebo	PARAM	Weight	N	n	AVAL	50
TRT01A	Drug A	PARAM	Weight	MEAN	Mean	AVAL	78.6
TRT01A	Drug A	PARAM	Weight	SD	Std Dev	AVAL	13.7
TRT01A	Drug A	PARAM	Weight	N	n	AVAL	58

Each row represents a single value on the table or figure output.

```
# Minimal example generating an ARD with {cards}
cards::ard_continuous(
  data = cards::ADSL,
  by = TRT01A,
  variables = AGE
)
```

```
## {cards} data frame: 24 x 10

##   group1 group1_level variable stat_name stat_label  stat
## 1 TRT01A    Placebo    AGE          N          N      86
## 2 TRT01A    Placebo    AGE        mean        Mean 75.209
## 3 TRT01A    Placebo    AGE          sd          SD   8.59
## 4 TRT01A    Placebo    AGE        median        Median  76
## 5 TRT01A    Placebo    AGE          p25          Q1   69
## 6 TRT01A    Placebo    AGE          p75          Q3   82
## 7 TRT01A    Placebo    AGE          min          Min   52
## 8 TRT01A    Placebo    AGE          max          Max   89
## 9 TRT01A  Xanomeli...    AGE          N          N   84
##10 TRT01A  Xanomeli...    AGE        mean        Mean 74.381

## i 14 more rows

## i Use 'print(n = ...)' to see more rows

## i 4 more variables: context, fmt_fun, warning, error
```

ARM (Analysis Results Metadata)

The Analysis Results Metadata (ARM) complements the ARD by describing how each result was derived. While ARD contains the results, ARM contains the instructions and definitions which defines how the result ARD was created.

Key Elements of ARM

ARM typically includes:

- **Output related information**
 - Information related to locations, standards, titles/footnotes.
 - **Variable meta**
 - Describes each column in the ARD
 - **Derivation methods**
 - This includes subsetting statements, population, p-value, Statistical methods (e.g., mean, logistic regression)
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Relationship Between ARD and ARM

- **ARD = Results data**
- **ARM = Metadata describing those results**

They work together to ensure: - **Traceability** → Results can be traced back to source data and methods
- **Reproducibility** → Analyses can be rerun using ARM definitions
- **Consistency** → Same definitions reused across outputs

In a complete ARS implementation: - ARD stores *what was calculated* - ARM explains *how and why it was calculated*

ARD in R with {cards}

The {cards} package produces ARD-like datasets where: - Each row is a result - Columns describe variables, groupings, and statistics

This aligns well with ARS principles and supports reproducible workflows in R.

2. Setup